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Professor Keegan

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Term Limits Influence Bill Introductions: Legislators Act Differently in Their Final Year

Most of us assume lawmakers' priorities remain steady throughout their careers—but in Colorado, term limits appear to spur dramatic changes in how legislators act. My research shows that lawmakers approaching the end of their terms introduce bills at rates that defy expectations—and it's more common for Democrats than Republicans.

To study how term limits influence legislative behavior, I used publicly available data from the Colorado General Assembly website. Each bill introduced into the 2016 to 2024 sessions was analyzed and information such as Number, Name, Summary, Subjects, Sponsors, Committees and Status was collected for the study. For example HB16-1005, “Residential Precipitation Collection”, described as “concerning the use of rain barrels to collect precipitation from a residential rooftop for nonpotable outdoor uses, on the topic of water. HB16-1005 was sponsored by Rep. Daneya Esgar, Rep. Jessie Danielson, and Sen. Michael Merrifield. Following introduction, this bill was assigned to the Agricultural, Livestock, & Natural Resources & Energy Committee¹. Finally, this bill became law. By examining patterns in bill sponsorship like this, I identified significant differences in legislative behavior for term-limited legislators compared to those not facing term limits in their current calendar year.

I was able to expand on previous research into web scraping, where previously I was able to obtain all the bills introduced into the 2024 session, to now look at every bill introduced from 2016-2024. Python libraries such as requests, pandas, and beautiful soup were critical for this process, not only obtaining the raw data, but then being able to parse and store the data into well labelled CSV files. Using the legislator database I also made a dataframe examining every legislator who served any time within the given time period with basic legislator information such as name, party, leadership information, year. Next, I did some basic subtraction to obtain a list of people who retired 2016-2023.² The study uncovered 71 legislators who retired within this time.

¹ Link to the final bill can be found [here](#) to verify the results of my web scraping analysis

² For example in the 2017 session, this would involve taking the 2016 senators and taking out the ones who appear again in 2017. By this logic, if a senator was present again in 2017 they must not have retired, and if they are not present they must have retired by 2016. However since our current term is 2024, I had to do research to find legislators who are retiring. Those retiring in 2024 are [Janet Buckner](#), [Chris Hansen](#) and [Kevin Van Winkle](#)

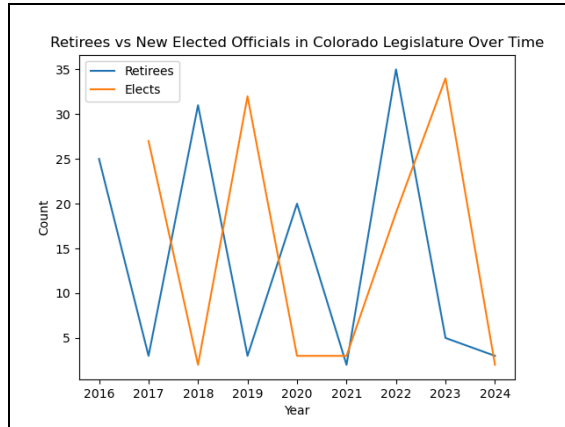


Fig 1: Graph showing the amount of retirees per session since 2016.

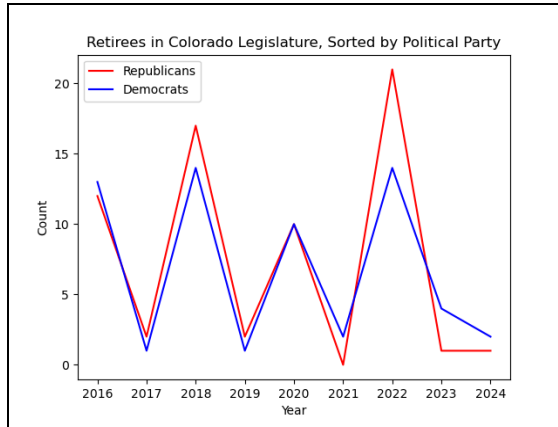


Fig 2: Further breaking up those who retired by political party. these lines tend to follow each other except in 2018/2022 cycle³

Next I would complete my statistical analysis and modelling. With my list of retirees, I wanted to investigate if their bill output was significantly different from the year which they would retire. Repeating the process for each legislator, I collected all the bills which they had sponsored the year before and year of retiring. I then tracked outcome information about the bills they sponsored, with those categories being N/A (usually referring to stalling or placed in a committee indefinitely), Became Law, Adopted, and Did not become law. This resulted in the following tables for each year, where each row represents a different representative.

	Retiree	Total Bills	Became Law	Adopted	Did Not Become Law	N/A
0	Galindo	0	0	0	0	0
1	Baumgardner	14	12	1	0	1
2	Lewis	7	0	1	0	6

Table 1: Table resulting from 2019 Cohort, from 2018 session.

	Retiree	Total Bills	Became Law	Adopted	Did Not Become Law	N/A
0	Galindo	12	11	0	0	1
1	Baumgardner	14	12	1	0	1
2	Lewis	22	8	2	0	12

Table 2: Table from the 2019 Cohort, from bills in the 2019 session.⁴

In figures one and two, we see the results of our data aggregation. To the naked eye, there are some obvious changes, such as Galindo who went from increased by twelve bills into session. Other changes seem imperceptible such as Baumgardner staying at 14 sponsorships in session.

In designing this study, I knew I wanted to include a chi squared test, but I wasn't sure where exactly I wanted to include it. After arriving at the 'expected' or normal amount of sponsorships per session contrasted with the 'observed' of how many bills sponsored in the final year of their term, I knew I wanted to include chi squared here. I employed a loop to go through the retirees, their observed/expected and spit out a chi squared + p-value per retiree, which was further broken into cohorts for 2017-2024 years. If it's been a while since you've taken a statistics course, chi squared is a statistical test which tests if the change between observed (bills sponsored in the year of retiring) and expected (bills normally sponsored) is statistically

³ Matplotlib library's plt.plot() was used to create plots in figures 1 and 2. Additionally plt.savefig() was used to save all plots.

⁴ dataframes' dfi.export(df_name, 'title.png') was used to download tables 1 and 2. Tutorial found [here](#)

significant or if it could be chance. Since we place the standard 95% confidence level, our alpha (which our p-value must be less than) is 0.05. It is important to establish these conditions prior to conducting any tests to our results are valid. Chi squared also requires >1 observation or expected, so I am also filtering out any cells which contain <1 observations or expected.

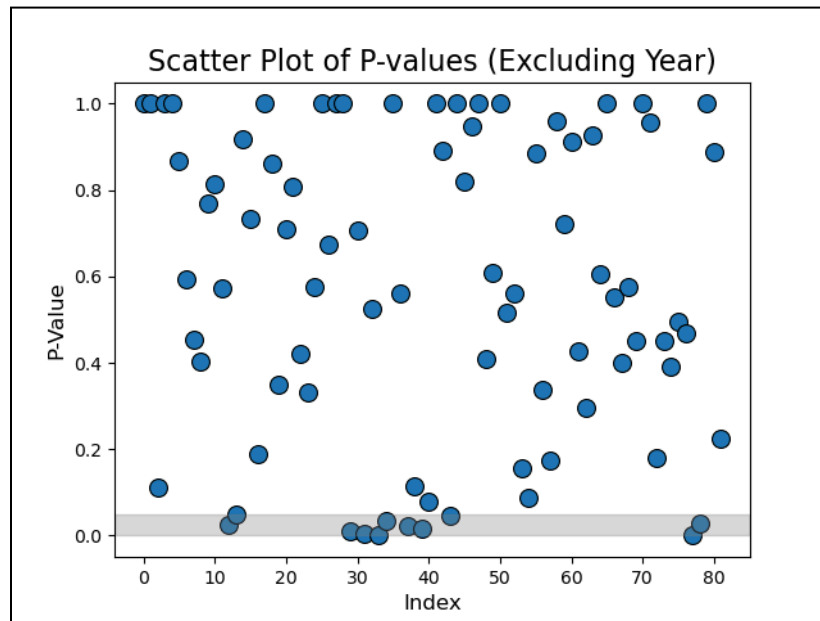


Fig 3: Plot showing every p-value the chi square analyses produced. P-values of 0.05 and below are shaded grey. Also interesting to note the large clump of p-values distributed at the 1.0 mark. P-values of 1 typically represent the least significant amount of change that can occur between the observed and expected variables. Seaborn's scatterplot function was used to create this plot. Additionally plt.savefig() was used to save all plots.

Aside from the fact that there are a few which have p-val's significant to us, plots like this aren't very helpful to us. We have no idea who or what the blue significant points represent. Thus I created the next plot to discuss those in depth.

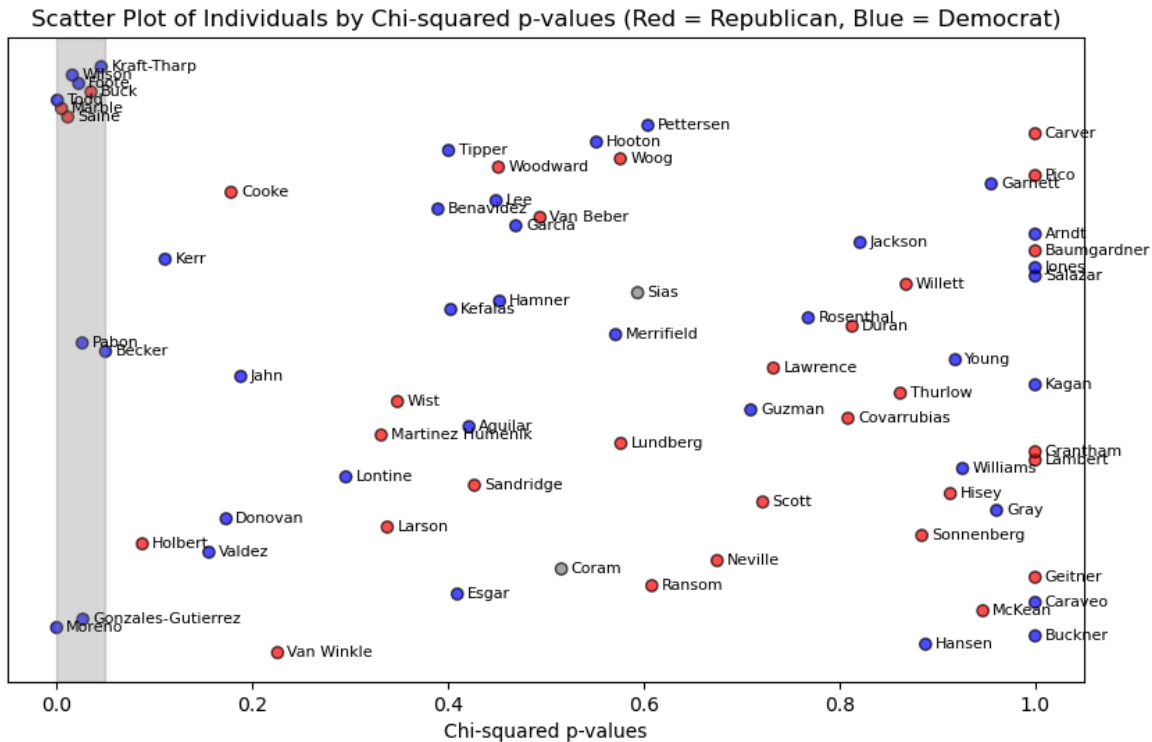


Fig 4: Showing the same plot but with our axes switched. Each point is labelled by corresponding party color, Red for Republicans and Blue for Democrats. Each point is labelled with the last name of the retiring legislator it represents. The part of the graph <0.05 is shaded grey to again, represent our “target” area. Matplotlib library scatterplot was used to generate this. Additionally plt.savefig() was used to save all plots.

Looking to the shaded area we see ten plotted points representing legislators with a p-val <0.05 . Respectfully those are Democrats Moreno, Gonzales-Gutierrez, Becker, Pabon, Todd, Wilson and Kraft-Tharpe. Of the Republicans those are Saine, Marble, and Buck. For anyone Paying attention that is ten total, seven of which are Democrats. 70% of those who statistically significantly changed the amount of bills they sponsored in their final year in office, are Democrats.

Of those who scored a p-val of 1.0 (quite literally the farthest away you can get from having a significant change) we have Democrats Buckner, Caraveo, Kagan, Salazar and Jones. We also have Republicans Geitner, Lambert, Grantham, Baumgardner, Pico and Carver. In percentages that is 54.5% (6/11) Republicans and 45.5% (5/11) Democrats. Across party lines, a majority of the p-val 1.0 legislators come from the 2018 cohort at 45.5% (5/11). The statistically significant legislators (scoring <0.05 majority came from the 2020 cohort at 80% (8/10).

Moving into our next analysis, I wanted to see if the variation between cohorts was due to chance or if it was statistically significant. Normally we would employ an Analysis of Variance ([ANOVA](#)) Test, but since our data isn’t distributed normally or had a similar variance to each other, we reach for it’s sister test: the [Kruskal Wallis](#). We can use the same p-val of 0.05 for this

test. After running it in [python](#) using the [scipy.stats](#) library and plotting the results of the cohorts as a box plot, we get the following:

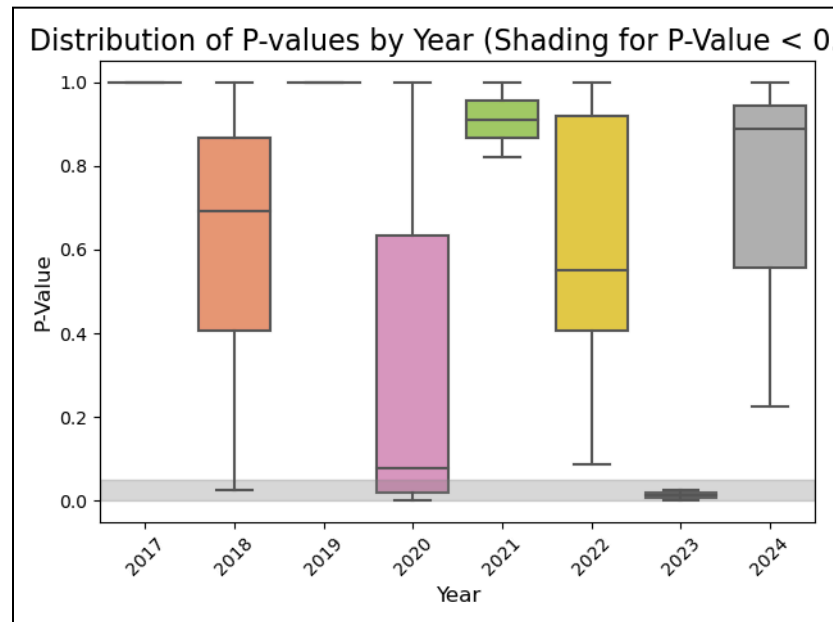


Fig 5: Plot showing the variation in p-val from the original test, sorted by year. This is a box and whisker plot. Again p-values below 0.05 are shaded grey to represent our “target” area. Usually in box and whisker plots, dots represent outliers in the data, in our case that would mean something was very wrong with the analysis, but their absence is a good indicator. [Quick refresher](#) is that around 75% of the data exists within the box, the center line represents the median, the exact middle of the data with the “whiskers” extending to the minimum and maximum values in the dataset. The bigger the box, the more spread our data is, smaller means the data is concentrated in one area. Seaborn’s box plot function was used to create this plot. Additionally `plt.savefig()` was used to save all plots.

Specifically, our test resulted in a test statistic of 19.6 and a p-val of 0.0065. In plain terms, this means there is a 6.5% chance of obtaining the test statistic value. More importantly though, this means since $0.0065 < 0.05$, we reject the null hypothesis that the variation in cohorts is completely due to chance. This means that statistically, there is variation in the cohorts which is not random.

Moving back to the cohorts though, we found that only 37.5% (¾) of the cohorts had at least one statistically significant p-value retiree. These years were, in chronological order, 2018, 2020 and 2023. Searching for answers, nationally 2018 was a midterm election year while 2020 was a presidential election year. This means that our sample size for these was larger at 2018_retirees having 31 people, and twenty_retirees having 20 people while the overall average length for our other cohort years was $\sim 9^5$. The outlier here being the 2023 cohort which had 5 people, with only 2 actually containing non-zero data.

```
[{'Retiree': 'Moreno',
  'Chi-Squared': 15.068578767123286,
  'P-Value': 0.00010367473844961803,
```

⁵ `statistics.mean([len(lst) for lst in [seventeen_retirees, nineteen_retirees, twenty_one_retirees, twenty_two_retirees, twenty_four_retirees]])` returns 9.2, statistics documentation found [here](#)

```

'Degrees of Freedom': 1,
'Expected Frequencies': array([[50.34482759,  9.65517241],
                                [22.65517241,  4.34482759]])),
{'Retiree': 'Gonzales-Gutierrez',
 'Chi-Squared': 4.896019198193111,
 'P-Value': 0.02691867979953044,
 'Degrees of Freedom': 1,
 'Expected Frequencies': array([[13.52941176,  6.47058824],
                                [ 9.47058824,  4.52941176]]))}]

```

Fig. 6: Python output for the chi squared function associated with 2023 cohort (twenty_three_chi_squared). Here we see that the P-value output is within our <0.05 alpha target value. Tutorial found [here](#) for the process of running chi square in a python cell. Python library scipy.stats was immensely helpful in creating the chi squared function and the Kruskal Wallis test cohort function.

Likely this can be described as the 2023 cohort coincidentally having two legislators (Moreno and Gozales-Gutierrez) who changed their behaviors significantly. But the size of the sample being so small for this cohort ($n=2$) makes the sample as a whole appear changed.

On the topic of limitations, this analysis has a few notable ones which need to be mentioned. Firstly is that this study only goes back to 2016, which skews our data in the face of modern trends and assumes that term limits are the baseline.⁶ This study had to be modified given how difficult it is to access data before 2016 on the legislative website. The other main limitation is how rare retiring is beyond term limits. This was evident in the few average being a little over 9 people retiring in non-significant cohorts.

Additionally in the future, I would like to expand the data to years before 2016 to get a bigger sample size. Our imposed time restraint constricts the ability to assess long term trends in legislative behavior influenced by term limits. Future research should prioritize obtaining and processing historical legislative data and running the same kinds of analyses on the data. Here, we could see opportunities to collaborate with legislature reporters such as the journalists who spoke to our class, historians/archivists, and leveraging optical character recognition (OCR) technologies.

⁶ The original aim of this study was to see the affect of term limits on legislative behaviors, given that the amendment for term limits was passed in 1991 and the difficulty of accessing pre-2016 data, the aim and research question had to be modified to be within the scope of the course.